

Using Next-Generation API Platforms to Build Microservices Architecture

The microservices architecture allows the development of software apps as discrete, loosely connected modules. This article focuses on the role of API gateways in microservices—particularly on how the Kong API gateway and the AWS API gateway can be used to build microservices.

Designing a scalable and fault tolerant application is one of the major responsibilities of a software architect. An application can be scaled easily when the services are stateless and loosely coupled. Otherwise, it will create multiple problems. Distributing services across servers makes the application fault tolerant. The above two factors decide the stability and reliability of an application when it is being accessed by multiple users. In earlier days, even a complex application would run as a single service in an operating system. If the process or system went down, an enterprise would lose its customers during peak use of services. To overcome the issues in scaling and to reduce the risks of running an application as a monolith, enterprises such as Netflix, eBay, Twitter and Amazon have introduced us to the world of microservices. This concept enables the scaling of a specific component of an application rather than scaling the entire application when there is a demand for just a specific service.



Microservices

Microservices have an architectural style that structures an application as loosely coupled services. This reduces friction in the continuous deployment and continuous integration of complex and large applications. There are many advantages of choosing a microservices architectural style over monolithic architecture when developing a

complex application. Complex applications that have many services, with each one having a dependency on another service, are really difficult to scale. It is better to design services that are not tightly coupled to each other.

Monolithic applications put all of their functionalities into a single process, whereas the microservices architecture puts each element of functionality into a separate service. So, if any service goes down in a monolithic application, the end users will be impacted more. But if the service is decoupled and distributed in different servers, then the impact is a lot less. Scaling an application through microservices is very easy.



Figure 1: Microservices architecture

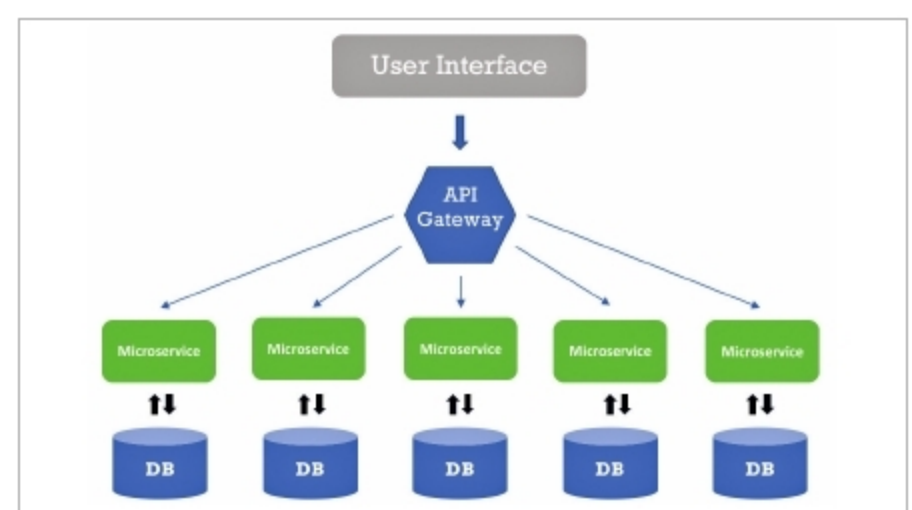


Figure 2: A typical microservices architectural style for designing an application